MySQL Cheetsheet by Kaniz Fatema

Here's a list of general SQL queries we've used in our lab so far.

SQL Quries & Relational Algebra

1. Database Operations

Operation	SQL Command
Create Database	`CREATE DATABASE DB_name;`
Delete Database	`DROP DATABASE DB_name;`

2. Table Operations

Operation	SQL Command
Create Table	`CREATE TABLE table_name (...);`
Insert Data Into Table	`INSERT INTO tbl_name VALUES (...);`
Rename Column Name	`ALTER TABLE tbl_name CHANGE old_col_name new_col_name new_data_type;`
Delete Column Data	`DELETE FROM tbl_name WHERE condition;`
Delete Row Data	`DELETE FROM tbl_name WHERE condition;`

3. Keys

Operation	SQL Command
Define Primary Key (create)	`CREATE TABLE table_name (col_name data_type, PRIMARY KEY(col_name));`
Add Primary Key (after)	`ALTER TABLE table_name ADD PRIMARY KEY (col_name);`
Define Composite Primary Key	`CREATE TABLE table_name (col1 data_type, col2 data_type, PRIMARY KEY (col1, col2));`
Add Composite Primary Key	`ALTER TABLE table_name ADD PRIMARY KEY (col1, col2);`
Drop Primary Key	`ALTER TABLE table_name DROP PRIMARY KEY;`

4. String Functions

Operation	SQL Command	Relational Algebra
Find string length	`SELECT LENGTH('Kaniz');`	Not represented in relational algebra
Get first 3 characters of a string	`SELECT SUBSTR('Kaniz', 1, 3);`	Not represented in relational algebra
Starts with (LIKE)	`SELECT * FROM table_name WHERE column_name LIKE 'A%';`	σ_{column_name LIKE 'A%'} (table_name)
Ends with (LIKE)	`SELECT * FROM table_name WHERE column_name LIKE '%Z';`	σ_{column_name LIKE '%Z'} (table_name)
Contains (LIKE)	`SELECT * FROM table_name WHERE column_name LIKE '%AI%';`	σ_{column_name LIKE '%AI%'} (table_name)

5. Filter Data

Operation	SQL Command	Relational Algebra
Filter using WHERE	`SELECT * FROM table_name WHERE column_name = 'value';`	σ_{column_name = 'value'} (table_name)
Filter with IN	`SELECT * FROM table_name WHERE column_name IN ('A', 'B', 'C');`	σ_{column_name ∈ {'A', 'B', 'C'}} (table_name)
Filter with NOT IN	`SELECT * FROM table_name WHERE column_name NOT IN ('X', 'Y');`	σ_{column_name ∉ {'X', 'Y'}} (table_name)
Filter with >, <, >=, <=	`SELECT * FROM table_name WHERE column_name > 100;`	σ_{column_name > 100} (table_name)
Filter with AND, OR	`SELECT * FROM table_name WHERE age > 18 AND city = 'Dhaka';`	σ_{age > 18 ∧ city = 'Dhaka'} (table_name)
Filter with BETWEEN	`SELECT * FROM table_name WHERE age BETWEEN 18 AND 30;`	σ_{age ≥ 18 ∧ age ≤ 30} (table_name)
Filter with LIKE	`SELECT * FROM table_name WHERE name LIKE 'A%';`	σ_{name LIKE 'A%'} (table_name) (expressed as selection)

6. COUNT Function

Operation	SQL Command	Relational Algebra
Count all rows in a table	`SELECT COUNT(*) FROM table_name;`	`G_{COUNT(*)}(table_name)`
Count non-null values in a column	`SELECT COUNT(column_name) FROM table_name;`	`G_{COUNT(column_name)}(table_name))`
Count with condition	`SELECT COUNT(*) FROM table_name WHERE age > 18;`	`G_{COUNT(*)}(σ_{age > 18}(table_name))`

7. Union and Intersection Operations

Operation	SQL Command Example	Relational Algebra Example
Union of tables	`SELECT * FROM table1 UNION SELECT * FROM table2;`	`table1 ∪ table2`
Union with condition	`SELECT * FROM table1 WHERE age > 18 UNION SELECT * FROM table2 WHERE age > 18;`	`σ_{age > 18}(table1) ∪ σ_{age > 18}(table2)`
Intersection of tables	`SELECT * FROM table1 INTERSECT SELECT * FROM table2;`	`table1 ∩ table2`
Intersection with condition	`SELECT * FROM table1 WHERE age > 18 INTERSECT SELECT * FROM table2 WHERE age > 18;`	`σ_{age > 18}(table1) ∩ σ_{age > 18}(table2)`

Join Operations

Operation	SQL Command Example	Relational Algebra Example
Natural Join	`SELECT * FROM table1 NATURAL JOIN table2;`	`π_column1, column2 (table1 □ table2)`
Inner Join	`SELECT * FROM table1 INNER JOIN table2 ;`	`π_column1, column2 (table1 □ table2)`
Left Outer Join	`SELECT * FROM table1 LEFT OUTER JOIN table2;`	`π_column1, column2 (table1 □ table2)`
Right Outer Join	`SELECT * FROM table1 RIGHT OUTER JOIN table2 ;`	`π_column1, column2 (table1 □ table2)`
Cartesian Product	`SELECT * FROM table1, table2 WHERE table1.id = table2.id;`	`π_{table1.id = table2.id}(table1 × table2)`

Projection Operation

Operation	SQL Command Example	Relational Algebra Example
Project specific columns	`SELECT column1, column2 FROM table_name;`	`π_{column1, column2}(table_name)`